Diversity You Cannot See From the Mean: Calibrated Detection of Latent Failure-Mode Structure in Open-Model Ecosystems

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Abstract

Mean-level diversity statistics for a panel of models, such as double-fault and pairwise disagreement, score behavioural spread from how often members fail, or answer, alike. We show that for failure outcomes this class of statistic is degenerate: mean pairwise co-failure, and with it mean pairwise disagreement, is a deterministic function of item difficulty alone and carries no information about shared behaviour once difficulty is accounted for. On 1,228–1,362 open language models across five benchmarks, the "excess" co-failure a naive computation reports is identically zero under the correct null. Calibrating against that null, the exact fit-free conditional distribution implied by Rasch sufficiency, reveals structure invisible to the mean in both directions: the raw participation ratio of 1.5–3.6, which reads as near-total redundancy, becomes 17–119 once calibrated, while a residual correlation of 2.9–11.9$\times$ its null remains and is concentrated in 4–12 latent behavioural dimensions rather than one shared factor. We then show the degeneracy is not merely an identity but reverses a decision: on a 2×2 grid of synthetic red-teaming suites with known ground truth, double-fault ranks a suite with planted shared failure modes as *more* diverse than an independent one in both difficulty regimes, and pairwise disagreement does so whenever probe potency is heterogeneous, while the calibrated statistic is correct in every cell. We describe the calibration as a drop-in diagnostic for diversity-aware benchmarks rather than a replacement for search-based diversity methods.

1 The measurement problem

A recurring move in classifier-ensemble diversity and, increasingly, in diversity- and coverage-aware safety evaluation is to summarise a panel's behavioural spread from pairwise (dis)agreement: how often do two members fail, or answer, the same way. This workshop's scope lists "diversity and coverage metrics for safety evaluation" and "behavioural repertoires of failures" as central objects. We show that the most common member of this family, mean pairwise co-failure, catalogued as the double-fault measure by Kuncheva & Whitaker (2003), is a deterministic function of item-level difficulty alone, and that mean pairwise disagreement inherits the same property.

Lemma 1 (Mean co-failure and mean disagreement depend only on margins)**.** *Let $F_{im} = 1$ iff panel member i fails item m, with $c_m = \sum_i F_{im}$ and $p_i = \frac{1}{M}\sum_m F_{im}$. The mean pairwise co-failure rate is $O = \frac{1}{MN(N-1)}\sum_m (c_m^2 - c_m)$, a function of the item margins $\{c_m\}$ alone; and the mean pairwise disagreement rate is $2\bar{p} - 2O$ with $\bar{p} = \frac{1}{N}\sum_i p_i = \frac{1}{NM}\sum_m c_m$, hence also a function of the item margins alone.*

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Benchmark	N	rms $\lvert R_{ij}\rvert$ obs.	null	ratio	participation ratio raw	calibrated	null	dims $>$ edge
ARC-Challenge	1362	0.2483	0.0449	5.5×	1.9	23.6	449.3 ± 3.4	6
Winogrande	1361	0.2691	0.0409	6.6×	3.6	20.1	493.8 ± 3.0	5
TruthfulQA	1334	0.2942	0.0583	5.1×	1.5	17.4	309.7 ± 2.4	5
GSM8K	1228	0.0987	0.0346	2.9×	2.1	118.6	553.1 ± 2.2	4
HellaSwag	1362	0.2457	0.0206	11.9×	1.5	27.3	958.0 ± 3.2	12

Table 1: Residual structure after exact-margin conditioning. "Raw" is the participation ratio of the unconditioned correlation matrix; "calibrated" is that of R. Null rms is the mean over 40 curveball replicates; null PR is mean $\pm$ SD over 60 independent-chain replicates with both margins exactly preserved. "Dims $>$ edge" counts eigenvalues of R exceeding the null's largest eigenvalue.

Proof. $\sum_{i\neq j} F_{im}F_{jm} = c_m^2 - c_m$ since $F_{im} \in \{0,1\}$; and pairwise disagreement is $p_i + p_j - 2C_{ij}$ with $C_{ij} = \frac{1}{M}\sum_m F_{im}F_{jm}$, whose mean over ordered pairs is $2\bar{p} - 2O$. □

Any randomization of which member fails which item that preserves each item's failure count therefore leaves both statistics exactly invariant, with zero variance. A benchmark with a few very hard items shows high mean co-failure even if every member fails independently given the item; a benchmark of uniformly moderate items shows low co-failure even for identical clones. The statistics conflate item difficulty with shared behaviour, and more data does not resolve this: it is an identity, not a sampling artifact. The same degeneracy is known in ecology (Schluter, 1984; Gotelli, 2000) and psychometrics (Cronbach, 1951).

2 A calibrated diversity signal, at scale

The fix is to condition on the item margins exactly rather than compare against an uninformative uniform baseline. The row and column margins of a binary outcome matrix are the jointly sufficient statistics of the Rasch model, so the uniform distribution over margin-matched matrices is the unique fit-free null that removes item difficulty without estimating anything; we sample it with curveball (Strona et al., 2014), verified against exhaustive enumeration on four small fibres (χ^2 uniformity not rejected, $p \in \{0.05, 0.23, 0.73, 0.15\}$). Let P be the Rasch mean field matching the margins, $D = (FF^\top - PP^\top)/M$ the residual co-failure matrix and R its unit-diagonal scaling; we report the rms off-diagonal $|R_{ij}|$, the participation ratio $\mathrm{PR} = (\sum_k \lambda_k^+)^2/\sum_k(\lambda_k^+)^2$ of R's positive spectrum, and the number of eigenvalues above the null's largest, each against exact-margin replicates. We apply this to the archived Open LLM Leaderboard: $1{,}228$–$1{,}362$ open models per benchmark, five benchmarks, harvested from result files at zero inference cost.

Table 1 makes the calibration gap concrete, and it runs in both directions. The raw participation ratio of 1.5–3.6 reads as near-total redundancy in the panel; that number is driven almost entirely by the single shared item-difficulty eigenvalue, and after conditioning the same statistic reads 17–119 (PR is a monotone transform of mean squared residual correlation, $\mathrm{PR} = N/(1 + (N-1)\overline{R_{ij}^2})$ for the unclipped spectrum, so we use it as a calibrated summary rather than a count of independent members). At the same time the calibrated value is only 2.8–21.4% of its own exact-margin null, so the panel is far more concentrated than a margin-matched population would be by chance, and the concentration is not one dominant shared factor: on ARC the leading eigenvalue carries 54% of squared residual mass (a single factor would carry $\approx 99\%$), and 4–12 eigenvalues exceed the null's spectral edge across the five benchmarks. Less redundant than the raw statistic suggests, more concentrated than chance, and multi-dimensional: none of the three is visible from a mean-level statistic, which by Lemma 1 cannot see any of it.

3 This is not just duplicate members

A natural objection to any concentration finding is that it is driven by near-duplicate panel members. Clustering models by pairwise agreement and keeping one representative per cluster at threshold 0.90 removes 900 of ARC's 1,362 models and moves the calibrated participation ratio from 23.6

Probe potency	Structure	Double-fault	Disagreement	rms ratio	dims > edge
heterogeneous	independent	0.3447	0.3092	1.04×	1
heterogeneous	shared modes	0.3184	0.3431	2.18×	4
homogeneous	independent	0.2624	0.4976	0.99×	0
homogeneous	shared modes	0.2561	0.4724	2.89×	4

Table 2: Synthetic suites with known ground truth. Lower double-fault and higher disagreement are conventionally read as more diverse. Within each difficulty regime the independent suite is the diverse one, so a correct metric puts it at lower double-fault and higher disagreement than the shared-modes suite in the row below it.

only to 29.0 (its null falls from 449.6 to 289.3). Recomputing every statistic at threshold 0.95, which discards 21–64% of each benchmark's models with the Rasch fit and null redrawn, moves the rms excess ratio by at most $1.14\times$ on every benchmark and leaves the eigenvalue count unchanged on three of five. Nor is ordinary discrimination heterogeneity among otherwise independent members sufficient: a two-parameter item-response simulation with $\mathrm{sd}(\log a) = 0.60$ tops out at rms residual correlation 0.076 against the observed 0.248 on ARC. What best matches the observed spectral signature ($\lambda_1^2 / \sum \lambda_k^2 = 0.540$) is a simulated population with two ability dimensions rather than one (0.457), not a population of duplicated members (0.102). The calibrated diagnostic is picking up genuine multi-dimensional behavioural structure, not redundant submissions or a single unmodelled discrimination parameter, though we cannot exclude that the extra dimensions are unmodelled abilities rather than shared failure modes.

4 Does the degeneracy change a decision?

Lemma 1 is an identity, and an identity does not by itself show that any decision goes wrong. We tested that directly, with the kill condition registered in advance: if the naive metrics rank the suites correctly, the degeneracy is not decision-relevant and the governance claim below must be withdrawn.

We generate suites of 200 members $\times$ 500 probes in the shape of a red-teaming evaluation, where $F_{im} = 1$ means probe m succeeded against member i, and cross two factors: probe potency heterogeneous (sd of logit difficulty 2.0) or near-homogeneous (0.15), and members either conditionally independent given the probe or drawn in five clusters with planted shared failure modes. Ground truth is known by construction: within a difficulty regime, the independent suite is the diverse one. A metric is correct if it scores that suite as more diverse.

Table 2 separates the two naive metrics. **Double-fault is wrong in both regimes**: it scores the shared-modes suite as more diverse than the independent one under heterogeneous potency (0.3184 against 0.3447) and again under homogeneous potency (0.2561 against 0.2624), because it tracks the overall failure rate that planting shared modes also changes. **Disagreement is wrong when potency is heterogeneous** (0.3431 for shared modes against 0.3092 for independent) and right when it is homogeneous (0.4976 against 0.4724), which is what Lemma 1 predicts: disagreement is $2\bar{p} - 2O$, so it inherits the degeneracy exactly and is rescued only when the margins happen not to vary. **The calibrated statistic is correct in all four cells**, scoring both independent suites at their own null ($1.04\times$ and $0.99\times$, with ≤ 1 eigenvalue above the edge) and both shared-modes suites well above it ($2.18\times$ and $2.89\times$, four eigenvalues above the edge). A search procedure scored by double-fault would prefer the wrong suite in either regime; one scored by disagreement would do so exactly when its probe suite has variable potency, which is the realistic case.

5 Relevance to diversity-aware benchmark governance

Section 4 shows the failure is not hypothetical, and this workshop's scope includes diversity-aware, continuously evolving safety benchmarks and coverage metrics for dynamic evaluation. The failure mode above is not specific to static multiple-choice accuracy: any *mean-level* coverage or diversity score computed from raw pairwise (dis)agreement over a set of test cases, whether attack variants, red-team probes or behavioural test suites, inherits the same degeneracy whenever test-case difficulty (in a safety setting, attack potency) varies across the suite. A search procedure that appears to be discovering diverse failure modes because raw disagreement is high may instead be finding items of heterogeneous difficulty; a procedure that looks converged because raw agreement is high may be sitting on a suite of uniform difficulty while covering genuinely distinct behavioural axes. Margin calibration, comparing against the exact conditional null implied by the suite's own per-item outcome counts rather than against a uniform baseline, is a domain-agnostic correction available wherever a diversity or coverage score is built from pairwise outcome comparisons, independent of whether the search that produced the suite is evolutionary, adversarial or manual. Its cost is one Rasch mean-field fit and a few dozen curveball replicates, and it requires no access to the models beyond their per-item outcomes.

6 Limitations

We study existing open models on static, single-turn multiple-choice benchmarks, not the multi-turn, tool-using or agentic settings central to this workshop, and we have not run the calibration inside an active evolutionary or quality-diversity search loop; that is the natural next step rather than something we report. The suites of Section 4 are synthetic and their generating process is the one the calibration assumes is wrong, so they establish that the naive metrics *can* invert a real ordering, not how often that happens in a deployed red-teaming pipeline. The conditioning model is unidimensional; Section 3 rules out discrimination heterogeneity and duplicate members as sufficient explanations but cannot separate genuine behavioural concentration from an underfit null family. The population is self-selected leaderboard submissions. Code, data and the pre-registration with dated amendments will be linked in the camera-ready. A large language model was used as a coding and drafting assistant; the authors specified and pre-registered the hypotheses, verified every derivation and number against executed-run artifacts, and are responsible for the content.

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